CELLPILOT FEASIBILITY STUDY

Kitting station, 20 s test take

Kitting station · recorded 2026-10-17 · 3 cameras · 0:20 · stored result · model gemini-4-argon · analysed Oct 17, 2026, 11:58 AM

A kitting station. The operator takes parts from three bins (A, B, C) and places one of each into the slots of a tray, orients the third part, compares the kit with a reference card, applies a label, and puts the finished kit into a bag on the out pile.

1 of 3 steps can be automated · 1 stays human · 1 needs a fixture

Engineer: 0 of 3 approved · 3 not yet signed off

Across two observed cycles, the operator positions a rectangular power bank on a flat wooden workstation, slides it across the tabletop to a marked target location, and lifts it into the air for manual handling and visual inspection. Sliding the rigid rectangular unit to the fixed tabletop marker is suitable for immediate automation with a 6-axis cobot such as a UR10e or Doosan M0609 equipped with a two-finger parallel gripper. However, initial part positioning on the table varies significantly between cycles and requires a locating fixture or input nest before a robot can reliably pick the part, while the final bimanual mid-air reorientation and visual check should remain with a human operator.

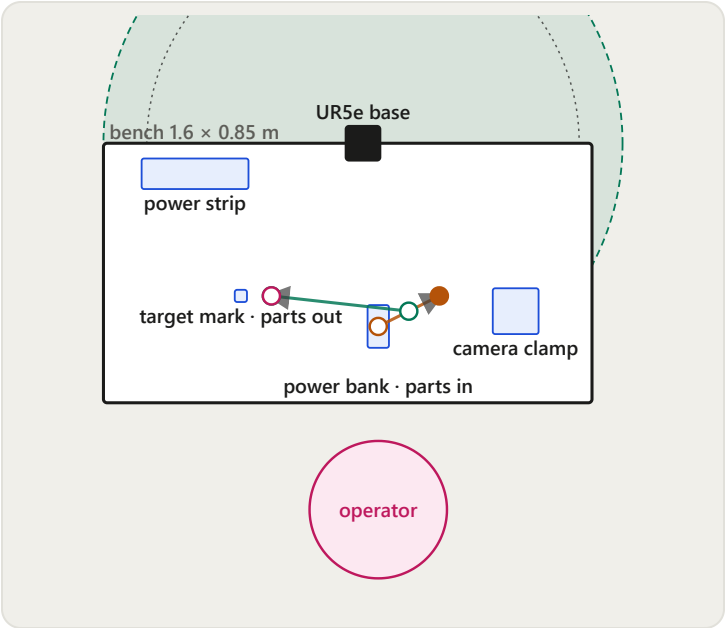
Steps, evidence and sign-off

Step and evidence	Verdict	Sign-off
1 · Position power bank on table 2 occurrences · mean 2.2 s · confidence 0.85 • The power bank sits loose in the middle of the bare tabletop with no mechanical stops or guide rails to establish a repeatable pickup pose. `cam1 at 00:08.0` • In the second cycle the operator places the power bank near the far edge of the table rather than the center, showing that a locating fixture is needed to present the part in a known position. `cam2 at 00:16.0` **Verifier:** Boundaries and verdict match what is seen across cam0, cam1, and cam2, as the power bank is placed in varying locations on the bare tabletop without a locating fixture. **Recommendation:** Install a mechanical nesting bracket or gravity-fed locating nest on the table so the incoming power bank arrives in a deterministic orientation for picking or sliding. Fixture: Mechanical corner-stop nest or passive locating tray with guide bevels.	NEEDS FIXTURE verifier agrees	not signed off

Step and evidence	Verdict	Sign-off
2 · Slide power bank to target mark 2 occurrences · mean 1.7 s · confidence 0.88 • The power bank has rigid parallel sides and is moved along an unobstructed flat surface to the marked target zone near the L sticker, which a two-finger parallel gripper can execute without tactile judgement. `cam0 at 00:10.0` • The operator translates the lightweight rectangular casing across the tabletop to the same destination area on the right side of the frame, well within the reach and payload of a Doosan M0609 or UR10e cobot. `cam1 at 00:18.0` **Verifier:** Start and end times in cam0, cam1, and cam2 align within one second, and translating the rigid rectangular casing to the marked target zone is well suited for a cobot with a parallel gripper. **Recommendation:** Automate with a UR5e equipped with a standard two-finger parallel gripper (Robotiq 2F-85 class) using compliant linear jogging along the tabletop to shift the 280 g power bank to the target mark.	AUTOMATE verifier agrees	not signed off
3 · Lift and inspect power bank 2 occurrences · mean 2 s · confidence 0.86 • The operator lifts the power bank high off the table and uses both hands in mid-air to reorient and inspect the front face of the device. `cam0 at 00:19.0` • The raised unit is transferred and stabilized between the operator's left and right hands during visual inspection, which cannot be replicated by a single two-finger parallel gripper. `cam1 at 00:19.0` **Verifier:** The lift and mid-air bimanual inspection intervals match all three camera views, and keeping this visual check and two-handed reorientation with a human operator is appropriate. **Recommendation:** Keep with the operator because mid-air bimanual reorientation and visual inspection cannot be replicated by a single parallel gripper without adding an expensive multi-camera vision system.	KEEP HUMAN verifier agrees	not signed off

Parts seen (estimated): power bank (~280 g, 140×70×20 mm)

Layout estimated from video



UR5e reach 850 mm needed reach 708 mm pick place

Bench 1.6 × 0.85 m; scale confidence 0.8. Scale is anchored by the 140x70 mm power bank, the operator's hand (~90 mm palm width), and shoulder width (~450 mm). The wooden dining table is approximately 1.6 m long by 0.85 m wide; the pick/place coordinates for s1 and s2 vary slightly between cycle 1 (moving right to left) and cycle 2 (moving left to right towards the target mark). Positions are the model's estimate from the three views, not measurements.

Robot

UR5e (Universal Robots) · passes the checks

Payload 5 kg · needs 1.28 kg ✓

Reach 850 mm · needs 708 mm ✓ · estimated layout

Price USD 35k–42k (indicative)

Arm	Payload	Reach	Price*	Verdict
UR3e Universal Robots	3 kg ✓	500 mm ✗	USD 23k–28k	reach short by 208 mm
xArm 6 UFACTORY	5 kg ✓	700 mm ✗	USD 10k–14k	reach short by 8 mm
UR5e Universal Robots	5 kg ✓	850 mm ✓	USD 35k–42k	**chosen**
Franka FR3 Franka Robotics	3 kg ✓	855 mm ✓	USD 20k–30k	also adequate; a larger arm than this job needs
Doosan M0609 Doosan Robotics	6 kg ✓	900 mm ✓	USD 30k–40k	also adequate; a larger arm than this job needs
UR10e Universal Robots	12.5 kg ✓	1,300 mm ✓	USD 50k–60k	also adequate; a larger arm than this job needs

Needed payload 1.28 kg: the heaviest of the parts in the automatable steps is power bank at about 280 g plus a 1 kg allowance for the gripper and fingers, assuming the rated payload is at the tool flange and excludes the gripper. Needed reach 708 mm: the farthest pick or place of the automatable steps from the proposed base, kept within 85% of the rated reach; the layout is estimated from video, so treat it as approximate. UR5e (5 kg, 850 mm) is the smallest

arm in the catalogue that passes every check, with 3.72 kg of payload margin; also adequate: Franka FR3, Doosan M0609, UR10e. Not proposed: UR3e (reach short by 208 mm); xArm 6 (reach short by 8 mm).

Cycle time

5 s per cycle today over 2 cycles; automatable share 33 %.



- 1 · Position power bank on table — 2.2 s · 44 %
- 2 · Slide power bank to target mark — 1.6 s · 33 %
- 3 · Lift and inspect power bank — 1.9 s · 39 %

With a cobot (estimate): robot 2.2 s on its steps (× 1.3), operator 4.2 s, side by side 4.2 s. Robot time = the operator's time on the automate steps x 1.3, a conservative factor for a cobot moving at collaborative speed; no robot motion was planned or simulated. Side by side = the longer of the robot's time and the operator's remaining time, the operator keeping the seconds no step covers.

Payback at the default inputs

Inputs	SGD 14/h · 2 × 8 h · 250 days · robot SGD 40,000 (indicative) × 3 · grant 50 % capped at SGD 30,000
Investment	SGD 120,000
Grant	− SGD 30,000
Savings a year	SGD 18,480
Payback	**≈ 58.4 months**

Investment = robot price × system multiple (a complete cell costs 3 to 5 × the robot: Automation World; Fraunhofer IPA). Grant: Singapore's Productivity Solutions Grant (Enterprise Singapore); check the current terms. Edit every input on the analysis page.

Cobot twin (simulated)

UR5e model, base at 0.86 × 0.77 m from the bench corner: every automatable point reachable, collision-free; robot 6.3 s on its steps, side by side 6.3 s.

Step	Point	Reachable	Error	Collision
Slide power bank to target mark	pick	yes	0 mm	none
Slide power bank to target mark	place	yes	0 mm	none

Parts list (indicative): UR5e cobot arm (Universal Robots) USD 35,000–USD 42,000; Two-finger parallel gripper (Robotiq 2F-85 class) with fingers USD 4,000–USD 6,500; Locating nest or presenter, custom (one per needs-fixture step) USD 800–USD 2,500; Pedestal or bench mount, cabling, control box placement USD 800–USD 2,000; hardware total USD 40,600–USD 53,000.

Kinematic twin in MuJoCo with the Menagerie arm model: the base is placed where every automatable point is within 85 % of the rated reach with the most margin; the tool goes pick, lift, place with inverse kinematics (mink); each configuration and the path between them are checked for contact with the bench, the objects and the fixtures; time = the longer of joint travel at 60°/s and tool travel at 0.25 m/s, plus 0.5 s to close or open the gripper. The part is welded to the gripper, not grasped; the gripper is a 160 mm tool; nothing here is a physics or safety validation.

Engineer's notes

The proposed UR5e cobot will handle sliding the power bank across the table into the target staging zone using a standard parallel gripper. The operator retains the bimanual lifting and visual quality inspection of the device casing. At an estimated net investment of 90,000 SGD, the expected payback period is 58.4 months based on modest cycle time reductions from 5.0 to 4.15 seconds. The primary caveat is that automatable scope is currently minimal, meaning economic justification is weak unless more tasks like fixturing and inspection are brought into the cell.

Risks

- **Poor economic return** · Automating only a 1.65-second sliding step yields minimal labor savings, leading to a long 58.4-month payback. Verify whether adjacent feeding or packing steps can be bundled before issuing purchase orders.
- **Part surface scratching** · Sliding the bare power bank across table surfaces and guide stops risks cosmetic scuffs on the housing. Test tabletop materials using Delrin or UHMW-PE sheets and soft silicone gripper pads before final cell build.
- **Loose part misalignment** · Without active sensing or fixed stops, slight variance in drop placement will cause gripper collisions or miss-picks. Validate that a passive mechanical nest repeatedly seats the 140x70x20 mm casing within 1 mm.

Next steps

1. Fabricate a rapid 3D-printed locating nest for the 140x70x20 mm power bank and verify manual loading repeatability.
2. Benchmark slide friction and surface wear on prototype power bank casings using polyurethane or silicone gripper fingers.
3. Obtain turnkey quotation packages for a UR5e arm, Robotiq 2F-85 gripper, and basic base framing.
4. Re-evaluate station workflow to determine if pre-kitting from bins A, B, and C can be integrated to improve project payback.

Before committing

- Is a 58.4-month payback acceptable to management for automating only the sliding task?
- Can the upstream placement of the power bank be mechanically fixed so step s1 can also be automated?
- What cosmetic scratch tolerances apply to the power bank exterior during sliding transfers?

Written by gemini-3.8-flash from the analysis, not from the video; the verdicts were not changed.

Analysis a-20261017-045342-a12a · model gemini-4-argon · 22,850 prompt and 13,147 output tokens · 5 min 9 s · Every verdict, time and reason comes from the model's stored answer; the robot choice, cycle and payback arithmetic are the app's code. Printed Oct 17, 2026, 07:55 PM.